

Chapter 1 INTRODUCTION

1.1. Need for smaller flow passages

1.2. Flow channel classification

1.3. Basic heat transfer and pressure drop considerations

1.4. Special demands of microscale biological applications

Chapter 1

INTRODUCTION

1.2. Need for smaller flow passages

Fluid flow inside channels is at the heart of many natural and man-made systems.

Heat and mass transfer is accomplished across the channel walls in biological systems, such as the brain, lungs, kidneys, intestines, blood vessels, etc., as well as in many man-made systems, such as heat exchangers, nuclear reactors, desalination units, air separation units, etc.

A channel serves to accomplish two objectives:

- (i) bring a fluid into intimate contact with the channel walls and
- (ii) bring fresh fluid to the walls and remove fluid away from the walls as the transport process is accomplished.

The rate of the transport (Heat Transfer-LK) process depends on the surface area, which varies with the diameter D for a circular tube,

whereas the flow rate ($V \times A$ -LK) depends on the cross-sectional area, which varies linearly with D^2 .

Thus, the tube surface area to volume ratio varies as $1/D$.

Clearly, as the diameter decreases, surface area to volume ratio increases.

In the human body, two of the most efficient heat and mass transfer processes occur inside the lung and the kidney (Böbrek-LK), with the flow channels approaching capillary dimensions of around $4\mu\text{m}$.

Figure 1.1 shows the ranges of channel dimensions employed in various systems.

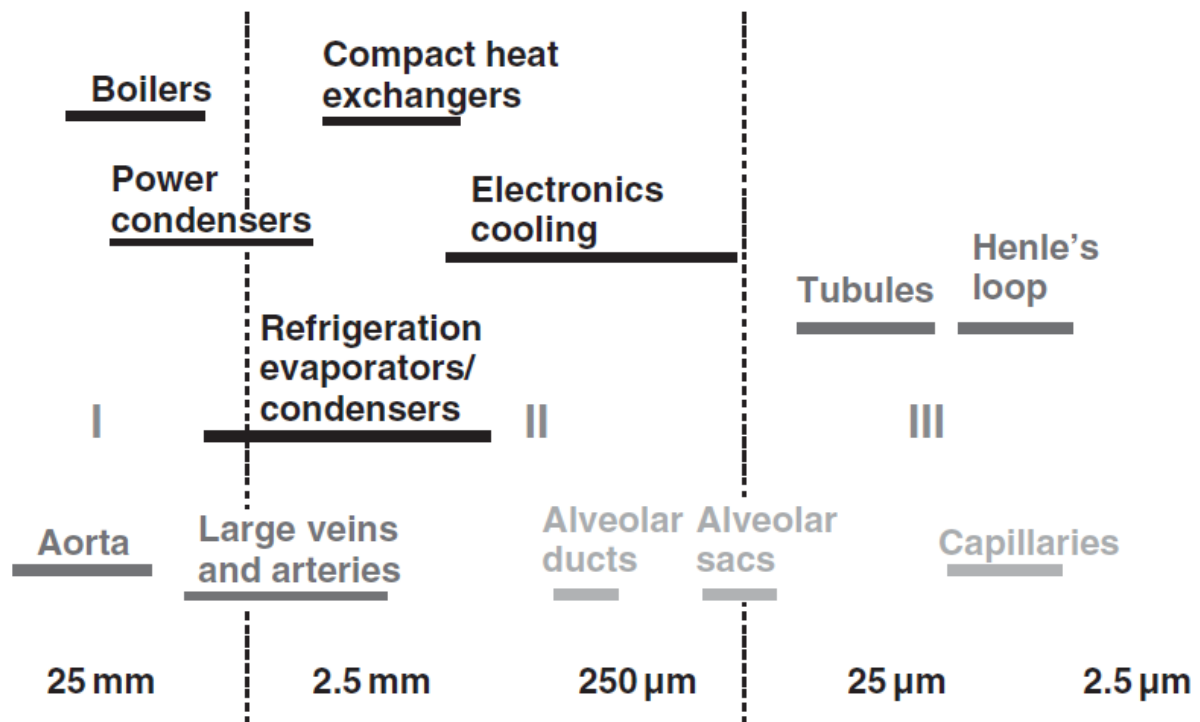


Fig. 1.1. Ranges of channel diameters employed in various applications, Kandlikar and Steinke (2003).

Since the dimensions of interest are in the range of a few tens or hundreds of micrometers, usage of the term “microscale” has become an accepted classifier for science and engineering associated with processes at this scale.

As the channel size becomes smaller, some of the conventional theories for (bulk) fluid, energy, and mass transport need to be revisited for validation.

Differences in modeling fluid flow in small diameter channels may arise as a result of:

(a) a change in the fundamental process, such as a deviation from the continuum assumption for gas flow, or an increased influence of some additional forces, such as electrokinetic forces, etc.;

(b) uncertainty regarding the applicability of empirical factors derived from experiments conducted at larger scales, such as entrance and exit loss coefficients for fluid flow in pipes, etc., or

(c) uncertainty in measurements at microscale, including geometrical dimensions and operating parameters.

In this book, the potential changes in fundamental processes are discussed in detail

1.2. Flow channel classification

The classification proposed by Mehendale *et al.* (2000) divided the range from

1 to 100 μm as microchannels,

100 μm to 1 mm as meso-channels,

1 to 6 mm as compact passages, and

greater than 6 mm as conventional passages.

Kandlikar and Grande (2003) considered the rarefaction effect of common gases at atmospheric pressure. Table 1.1 shows the ranges of channel dimensions that would fall under different flow types.

Table 1.1

Channel dimensions for different types of flow for gases at one atmospheric pressure.

Gas	Channel dimensions (μm)			
	Continuum flow	Slip flow	Transition flow	Free molecular flow
Air	>67	0.67–67	0.0067–0.67	<0.0067
Helium	>194	1.94–194	0.0194–1.94	<0.0194
Hydrogen	>123	1.23–123	0.0123–1.23	<0.0123

A more general scheme based on the smallest channel dimension is presented in Table 1.2.

Table 1.2

Channel classification scheme.

Conventional channels	$>3 \text{ mm}$
Minichannels	$3 \text{ mm} \geq D > 200 \mu\text{m}$
Microchannels	$200 \mu\text{m} \geq D > 10 \mu\text{m}$
Transitional Microchannels	$10 \mu\text{m} \geq D > 1 \mu\text{m}$
Transitional Nanochannels	$1 \mu\text{m} \geq D > 0.1 \mu\text{m}$
Nanochannels	$0.1 \mu\text{m} \geq D$

D : smallest channel dimension

In Table 1.2, D is the channel diameter.

In the case of non-circular channels, it is recommended that the minimum channel dimension; for example, the short side of a rectangular cross-section should be used in place of the diameter D .

We will use the above classification scheme for defining minichannels and microchannels.

1.4. Basic heat transfer and pressure drop considerations

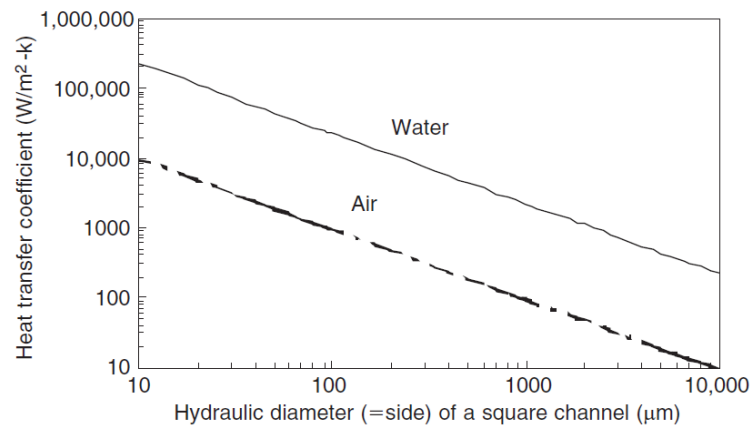


Fig. 1.2. Variation of the heat transfer coefficient with channel size for fully developed laminar flow of air and water.

For square channel, constant heat flux, fully developed laminar flow

$$Nu=3.61 \quad h = Nu \frac{k}{D_h}$$

The dramatic enhancement in h with a reduction in channel size is clearly demonstrated.

Channel size $\searrow$ heat transfer coefficient $\nearrow$

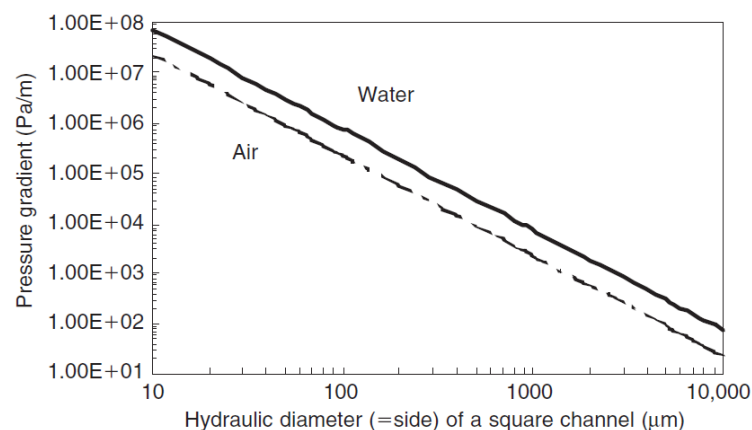


Fig. 1.3. Variation of pressure gradient with channel size for fully developed laminar flow of air and water. Mass flux $G = 200 \text{ kg/m}^2 \text{ s}$

Channel size $\searrow$ Pressure gradient $\nearrow$

1.5. Special demands of microscale biological applications

The use of minichannels, microchannels, and microfluidics in general is becoming increasingly important to the biomedical community. However, the transport and manipulation of living cells and biological macromolecules place increasingly critical demands on maintaining system conditions within acceptable ranges. For instance, human cells require an environment of 37°C and a pH=7.4 to ensure their continued viability. If these parameter values stray more than 10% then cell death will result.

1.5. Summary

Microchannels are found in many biological systems where extremely efficient heat and mass transfer processes occur, such as lungs and kidneys.

The channel size classification is based on observations of many different processes, but its use is mainly in arriving at a common terminology.

The heat and mass transfer rates in small diameter channels are also associated with a high pressure drop penalty for fluid flow. The fundamentals of microchannels and minichannels in various applications are presented in this book.

1.6. Practice problems

Problem 1.1 Calculate the heat and mass transfer coefficients for airflow in a human lung at various branches. Assume fully developed laminar flow conditions. Comment on the differences between this idealized case and the realistic flow conditions.

Hint: Consult a basic anatomy book for the dimensions of the airflow passages and flow rates, and a heat and mass transfer book for laminar fully developed Nusselt number and Sherwood number in a circular tube.

Problem 1.2 Calculate the Knudsen number for flow of helium flowing at a pressure of 1 millitorr (1 torr = 1 mmHg) in 0.1-, 1-, 10-, and 100-mm diameter tubes. What type of flow model is applicable for each case?

Hint: Knudsen number is given by $Kn = \lambda/D_h$, and the mean free path λ is given by $\lambda = (\mu\sqrt{\pi}/\rho\sqrt{2RT})$, where μ is the fluid viscosity, ρ is the fluid density, T is the absolute temperature, and R is the gas constant.

Problem 1.3 Calculate the pressure drop for flow of water in a 15-mm long 100- μm circular microchannel flowing at a temperature of 300 K and with a flow Reynolds number of (a) 10, (b) 100, and (c) 1000. Also calculate the corresponding water flow rates in kg/s and ml/min.